

Faculty of Engineering

Final Year Project Work Guidelines

**B.E. (Information Technology), 2019 Course (With
effect from Academic Year 2025 - 26)**



SAVITRIBAI PHULE PUNE UNIVERSITY

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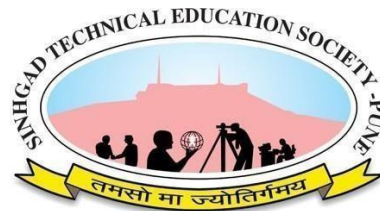
Sinhgad College of Engineering, Pune

Department of Information Technology

Final Year Project Work

Academic Year: 2025-2026

Project Title	:	AgroMate: Smart Insights for Better Farming
Project Area	:	Web Development and Machine Learning
Internal Guide	:	Prof. M. S. Bhosle



Sinhgad Institutes

A PROJECT REPORT ON

“AgroMate: Smart Insights for Better Farming”

A PROJECT REPORT ON

AgroMate: Smart Insights for Better Farming

SUBMITTED TO THE SAVITRIBAI PHULE PUNE UNIVERSITY, PUNE
IN THE PARTIAL FULFILLMENT FOR THE AWARD OF THE DEGREE

OF

BACHELOR OF ENGINEERING

IN

INFORMATION TECHNOLOGY

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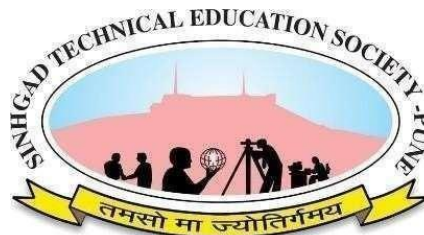
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CERTIFICATE

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is a bonafide work carried out by them under the supervision of Prof. M. S. Bhosle and it is approved for the partial fulfillment of the requirement of Savitribai Phule Pune University for the award of the Degree of Bachelor of Engineering (Information Technology)

This project report has not been earlier submitted to any other Institute or University for the award of any degree or diploma.

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Sneha Vijay Jadhav
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Abstract

AgroMate: Smart Insights for Better Farming is an intelligent agriculture platform designed to assist farmers in making data-driven decisions using machine learning (ML) and deep learning (DL) techniques. The system provides personalized recommendations and insights through its key modules: crop recommendation based on weather and soil data, plant disease detection using image analysis, optimized irrigation suggestions, and a digital mandi for secure produce transactions. AgroMate integrates these advanced models into a user-friendly web interface, enabling farmers to access critical information and predictive analytics easily. By harnessing the power of ML and DL, AgroMate aims to enhance crop yield, reduce risks, and promote efficient farming practices, paving the way for a more resilient and informed agricultural community.

Keywords: Machine Learning, Deep Learning, CNN, Image Classification, Real-Time Market Price, API Integration.

CHAPTER 1: INTRODUCTION

1.1 Introduction

Agriculture has always been a fundamental sector contributing to the economic development and sustainability of a nation. In countries like India, a significant portion of the population depends on agriculture for their livelihood. Despite its importance, the agricultural sector continues to face numerous challenges due to the continued reliance on traditional farming techniques. These methods often lack scientific backing and depend heavily on the farmer's experience, leading to inconsistent results and inefficiencies in production.

The increasing demand for food due to population growth has placed additional pressure on the agricultural sector to improve productivity and efficiency. However, factors such as improper crop selection, lack of awareness about soil health, delayed identification of plant diseases, and fluctuating market prices hinder farmers from achieving optimal outcomes. These challenges highlight the need for a modern, technology-driven approach to agriculture.

AgroMate is developed as a smart agricultural system that integrates machine learning and image processing techniques to assist farmers in making informed decisions. The system provides recommendations for suitable crops based on soil conditions, detects plant diseases through image analysis, and offers market insights to improve profitability. By digitizing farming processes and providing accurate, real-time information, AgroMate aims to enhance productivity and promote sustainable agricultural practices.

1.2 Background of the Project

The agricultural sector has undergone significant transformations over the years, but many farmers still rely on outdated practices due to limited access to modern technologies. Soil health plays a crucial role in determining crop productivity, yet many farmers are unaware of essential soil parameters such as nutrient levels and pH balance. This lack of awareness often leads to poor crop selection and reduced yield.

Another major challenge is the early detection of plant diseases. Traditional methods of disease identification are manual and require expertise, which may not always be available to farmers. Delayed detection can result in severe crop damage and financial loss. Additionally, market price fluctuations create uncertainty, making it difficult for farmers to decide the best time to sell their produce.

With the advancement of machine learning and image processing, it is now possible to analyze large datasets and extract meaningful insights. These technologies can be effectively utilized

to provide accurate crop recommendations, detect diseases early, and analyze market trends. AgroMate leverages these capabilities to create a unified system that addresses multiple agricultural challenges in a practical and efficient manner.

1.3 Relevance of the Project

The relevance of the AgroMate project lies in its ability to address real-world agricultural problems through technological innovation. The system provides a scientific approach to farming by replacing guesswork with data-driven decision-making. This not only improves crop productivity but also reduces the risk of failure.

The integration of disease detection using image processing ensures that farmers can identify and treat plant diseases at an early stage, minimizing damage and improving crop health. Market price analysis helps farmers make better financial decisions, thereby increasing profitability. Additionally, the digital logbook feature promotes organized record-keeping, which is essential for long-term planning and improvement.

In the context of increasing demand for food and the need for sustainable farming practices, AgroMate serves as a valuable tool for modernizing agriculture and supporting farmers in achieving better outcomes.

1.4 Aim and Goal

1.4.1 Aim

1. To develop a crop recommendation system that suggests the most suitable crops based on soil conditions.
2. To design a crop market price prediction model for forecasting future prices to help farmers make better selling decisions.
3. To implement a disease detection system using plant leaf images with accurate classification and pesticide suggestions based on identified diseases for effective and timely treatment.
4. To create a digital logbook for maintaining structured records of crop activities like irrigation, and pesticide usage.

1.4.2 Goal

1. **Efficient Crop Recommendation System** – Develops a reliable system that analyzes soil nutrients and environmental factors to suggest the most suitable crops, helping farmers improve yield and resource utilization.

2. **Reliable Market Price Prediction** – Provides accurate forecasting of crop prices using analytical models, enabling farmers to make better selling decisions and increase profitability.
3. **Accurate Disease Detection and Treatment Suggestion** – Implements an image-based detection system that identifies plant diseases at an early stage and recommends appropriate pesticides for effective crop protection.
4. **Smart Digital Record Management** – Establishes a digital logbook that stores and organizes farming activities such as irrigation and pesticide usage, supporting better monitoring and long-term agricultural planning.

1.5 Project Execution

The execution of the AgroMate project involves several stages, beginning with the identification of key agricultural challenges and the collection of relevant data. Soil data, crop information, and plant disease images are gathered from various sources to train machine learning models. The data is then preprocessed to ensure accuracy and consistency.

Machine learning algorithms are applied to develop models for crop recommendation and market analysis, while image processing techniques are used to build a disease detection system. These models are integrated into a web-based application using a structured architecture that separates the frontend, backend, and database components.

The system undergoes rigorous testing to ensure reliability, accuracy, and performance. User feedback is incorporated to improve usability and functionality. Finally, the project is documented in detail, providing insights into the design, implementation, and evaluation of the system.

CHAPTER 2: LITERATURE REVIEW

Sr No	Publication Details	Seed Idea	Drawbacks
1	Navarro et al. (2020), Comprehensive Review of IoT Solutions for Smart Farming.	Presented IoT-based smart farming systems integrating sensors, cloud computing, and AI for crop monitoring and decision-making.	Most systems focus on data collection but lack fully integrated platforms offering actionable, multi-aspect guidance.
2	Patil et al. (2025), IoT-Based Smart Irrigation and Soil Monitoring System.	Developed an IoT-based system that uses soil moisture and weather data for automated irrigation, improving water efficiency.	Tested only on small farms; faces challenges in large-scale implementation and maintaining connectivity in rural areas.
3	Frimpong et al. (2023), Strategies for Water-Smart Farming.	Reviewed cropping system designs that conserve water while maintaining crop yield using decision-support tools.	Existing tools do not fully consider local environmental variations, reducing real-world effectiveness.
4	Kumar et al. (2024), IoT Applications in Agriculture	Examined the use of IoT sensor networks for environmental data collection and advocated integration with AI for analysis.	Lacks unified systems combining IoT data with AI modules for disease detection and market prediction.

CHAPTER 3: AIM AND OBJECTIVES

3.1 Aim

The primary aim of the AgroMate system is to develop an intelligent and integrated agricultural platform that leverages machine learning and image processing techniques to assist farmers in making accurate and data-driven decisions. The system is designed to provide reliable recommendations for crop selection, enable early detection of plant diseases, and offer meaningful market insights. By combining these functionalities into a single platform, the project aims to enhance productivity, reduce uncertainty in farming practices, and promote sustainable agricultural development.

3.2 Objectives

1. Develop a Crop Recommendation System:

The objective of this module is to design and implement an intelligent crop recommendation system that suggests the most suitable crops based on soil parameters such as nitrogen, phosphorus, potassium, and pH levels. The system utilizes machine learning techniques to analyze these inputs and generate accurate recommendations. This helps farmers make informed decisions regarding crop selection, thereby improving productivity and efficient utilization of resources.

2. Design a Market Price Prediction Module:

The objective of this component is to develop a predictive system that analyzes historical and current market data to forecast future crop prices. By identifying trends and fluctuations in the market, the system assists farmers in determining the optimal time to sell their produce. This enables better financial planning and helps in reducing losses caused by unpredictable market conditions.

3. Implement Disease Detection Using Image Processing:

The objective of this module is to develop an image-based disease detection system that allows users to upload plant leaf images for analysis. The system applies machine learning or deep learning techniques to identify diseases accurately and classify them effectively. Based on the detected disease, appropriate pesticide or treatment suggestions are provided to ensure timely intervention and reduce crop damage.

5. Develop a Digital Logbook System:

The objective of this feature is to create a structured digital logbook that enables farmers to record and manage daily farming activities such as irrigation, fertilizer application, and pesticide usage. This system helps in maintaining organized records, which can be used for future analysis, better planning, and improvement of farming practices over time.

5. Enhance System Performance and Usability:

The objective of this aspect is to ensure that the system performs efficiently with high accuracy and minimal response time. The application is designed to be user-friendly and accessible, allowing farmers with limited technical knowledge to easily interact with the system. This improves usability and encourages wider adoption of the technology.

6. Ensure Data Security and System Reliability:

The objective of this component is to implement secure authentication mechanisms and data protection measures to safeguard user information. The system ensures confidentiality, integrity, and reliable storage of data, thereby providing a secure and trustworthy platform for users.

CHAPTER 4: METHODOLOGY

4.1 Problem Statement

Agriculture remains one of the most essential sectors for economic development and food security, yet it continues to face numerous challenges due to the lack of technological integration. Farmers often rely on traditional farming practices and personal experience rather than data-driven approaches, which leads to inefficiencies and reduced productivity. One of the major issues is the lack of proper knowledge about soil conditions, including nutrient levels such as nitrogen, phosphorus, potassium, and pH values. This often results in improper crop selection, leading to poor yield and soil degradation.

Another significant challenge is the early detection of plant diseases. In many cases, farmers are unable to identify diseases at an early stage due to limited expertise and lack of access to advanced diagnostic tools. This delay in detection can cause severe damage to crops and lead to financial losses. Additionally, the fluctuation of market prices creates uncertainty, making it difficult for farmers to decide the optimal time to sell their produce. The absence of reliable market insights further increases the risk of economic instability.

Moreover, there is a lack of structured record-keeping in traditional farming practices. Farmers often do not maintain systematic records of their activities such as irrigation schedules, fertilizer usage, and pesticide application. This makes it difficult to analyze past performance and improve future decision-making.

To address these challenges, there is a need for an integrated system that can provide accurate crop recommendations, detect plant diseases at an early stage, offer market insights, and maintain digital records of farming activities. The AgroMate system is designed to fulfill these requirements by leveraging machine learning, image processing, and web technologies to provide a comprehensive and intelligent solution for modern agriculture.

4.2 Overall Architecture

The AgroMate system is designed using a three-tier web application architecture, which ensures modularity, scalability, and efficient interaction between different components of the system. This architecture separates the application into three main layers, namely the client layer, server layer, and data and storage layer. Each layer performs a specific function and communicates with the others to ensure smooth operation of the system.

1. Client Layer (Frontend):

The client layer represents the user interface of the system, which is accessed through a web browser. It provides a responsive and user-friendly interface that allows farmers to interact with the application easily. This layer handles all user interactions such as form submission, image uploads, and navigation across different modules of the system. The frontend communicates with the backend server through HTTP requests, ensuring seamless data exchange and real-time response generation.

2. Server Layer (Backend - Flask Application):

The server layer acts as the core processing unit of the system and is implemented using a Flask-based web application. It is responsible for handling all incoming requests from the client layer and routing them to the appropriate modules for processing. The Flask application serves as a central controller that manages communication between different services and ensures proper execution of system functionalities.

The authentication module within this layer is responsible for validating user credentials during login and registration processes. It ensures secure access to the system by implementing authentication and authorization mechanisms. Session management is handled using token-based techniques, which help maintain user sessions and restrict access to authorized users only.

The core services layer contains the main business logic of the system. It includes the prediction service, which is responsible for crop recommendation using machine learning models such as Random Forest. The insights service provides valuable information related to market prices, weather conditions, and farmer-specific analytics. The artificial intelligence service is used for disease detection through advanced deep learning models and also supports chatbot functionality for user assistance and guidance.

3. Data and Storage Layer:

The data and storage layer is responsible for managing all persistent data required by the system. It uses a relational database to store user information, login credentials, prediction history, and farmer profiles. This ensures that all user-related data is securely stored and can be retrieved whenever required.

In addition to the database, this layer also includes pre-trained machine learning and deep learning models. These models are used for performing tasks such as crop recommendation and disease detection. By storing these models separately, the system ensures efficient execution and easy updates whenever improvements are made.

4. External Integration:

The AgroMate system integrates with various external services to enhance its functionality and provide real-time data. Weather APIs are used to fetch current and forecasted weather information, which is essential for agricultural decision-making. Market price databases provide updated pricing information for crops, helping farmers plan their sales effectively. Translation services are integrated to support multiple languages, making the system accessible to a wider audience. Additionally, chatbot APIs are used to provide interactive assistance and improve user experience.

4.3 Technology Stack

The AgroMate system is developed using a combination of modern technologies to ensure efficient performance, scalability, and ease of development. The technology stack is categorized based on different components of the system as follows:

1. **Web Framework (Backend):** Flask is used as the backend web framework for developing the application. It handles routing, request processing, and integration of machine learning models.
2. **Database Management:** SQLite is used for storing user data, prediction history, and farming records, while SQLAlchemy is used for database interaction.
3. **Machine Learning and Deep Learning Libraries:** PyTorch and TensorFlow/Keras are used for developing models for crop recommendation and disease detection.
4. **Data Processing and Feature Engineering:** Scikit-learn and Pandas are used for data preprocessing, analysis, and model training.
5. **Image Processing:** PIL and Pillow are used for handling and processing plant leaf images for disease detection.
6. **API Integration:** Requests library is used to connect external APIs for weather, market prices, and other services.
7. **Natural Language Processing and Translation:** Google Translate API and MyMemory API are used for multilingual support.
8. **Containerization and Deployment:** Docker is used for packaging the application and ensuring consistent deployment.
9. **Cloud Deployment:** Azure Container Apps is used for hosting the application on the cloud.
10. **Web Server:** Gunicorn is used as the production server for handling multiple user requests efficiently.

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4.4 Development Methodology

The AgroMate project follows an Agile Development approach with iterative cycles to ensure flexibility, continuous improvement, and effective implementation of system features.

1. **Requirements Analysis:** Identified the key challenges faced by farmers such as improper crop selection, lack of disease detection, and market uncertainty through surveys, research, and analysis of real-world agricultural problems.
2. **System Design:** Designed a modular and scalable system architecture based on a three-tier structure, enabling independent development of components such as crop recommendation, disease detection, and market analysis.
3. **Model Training:** Trained machine learning and deep learning models using diverse datasets such as crop recommendation datasets, PlantVillage dataset for disease detection, and weather datasets for environmental analysis.
4. **Implementation:** Developed the system using a full-stack approach, integrating frontend and backend components along with machine learning models, and ensured continuous integration for smooth functionality.
5. **Testing:** Performed various levels of testing including unit testing, integration testing, and user acceptance testing to ensure system accuracy, reliability, and proper performance.
6. **Deployment:** Deployed the application using containerization tools like Docker to ensure portability, scalability, and readiness for cloud-based environments.
7. **Evaluation:** Evaluated the system performance through testing and user feedback to validate accuracy, usability, and effectiveness in solving real-world agricultural problems.

CHAPTER 5: REQUIREMENT SPECIFICATION

5.1 Software Requirements

- Operating System : Windows 10/11 or above
- Web Framework : Flask ($\geq 3.0.0$)
- Coding Language : Python
- Database : SQLite with SQLAlchemy (3.1.1)
- Machine Learning Libraries : PyTorch (2.2.0), TensorFlow/Keras (2.8.0)
- Data Processing Tools : Scikit-learn (1.4.0), Pandas (2.2.0)
- Image Processing : PIL, Pillow (10.2.0)
- Containerization Tool : Docker (Latest Version)
- Cloud Platform : Azure Container Apps
- Web Server : Gunicorn (21.2.0)
- IDE : Visual Studio Code
- Web Browser : Google Chrome

5.2 Hardware Requirements

- Processor : Intel i5 or equivalent
- Hard Disk : 500 GB (Minimum)
- RAM : 8 GB (Minimum)
- Monitor : 15-inch or above
- System Type : 64-bit Computer System
- Input Devices : Keyboard and Mouse
- Internet Connection : Required for API access and deployment

CHAPTER 6: SYSTEM DESIGN

6.1 Design Details

1. Modular System Design:

The AgroMate system is designed using a modular approach where the application is divided into different modules such as crop recommendation, disease detection, market analysis, and digital logbook. Each module performs a specific function, which improves system organization and clarity. This structure allows easy development and testing of individual components.

The modular design also enhances scalability and maintainability. New features can be added without affecting the entire system, and errors can be easily identified and fixed within specific modules. This makes the system flexible and suitable for future enhancements.

2. Client-Server Architecture:

The system follows a client-server architecture where the frontend acts as the client and communicates with the backend server. The frontend is developed using HTML, CSS, and JavaScript, providing a simple and user-friendly interface for farmers.

The backend is implemented using Flask, which handles all processing tasks such as request handling, data processing, and model integration. This separation improves performance and allows independent development of frontend and backend components.

3. Integration of Machine Learning Models:

AgroMate integrates machine learning and deep learning models to provide intelligent outputs. The crop recommendation module uses a Random Forest model to suggest suitable crops based on soil parameters. The disease detection module uses a deep learning model to analyze leaf images and identify diseases. These models are integrated into the backend, enabling fast and accurate predictions for users.

5. Use of UML Diagrams:

UML diagrams are used to represent the system structure and workflow. These include Use Case, Activity, Sequence, and Class diagrams, which help in understanding system functionality and interactions. These diagrams provide clear visualization of data flow and communication between components, making system design easier to understand and implement.

6.2 System Architecture

1. Client Layer (Frontend):

The client layer provides the user interface through which farmers interact with the system. It includes features such as login, data input, image upload, and result display. The interface is designed to be simple and easy to use. The frontend communicates with the backend using HTTP requests, ensuring smooth data exchange and real-time response.

2. Server Layer (Backend - Flask Application):

The server layer is implemented using Flask and acts as the core processing unit. It receives user requests, processes them, and sends responses back to the client. This layer includes modules such as authentication, prediction services, and AI services for disease detection. It ensures efficient execution of system functionalities.

3. Data and Storage Layer:

This layer stores all system data including user information, prediction history, and farming records in a database. It ensures secure and organized data storage. Pre-trained machine learning models are also stored here and used by the backend for making predictions.

4. External Integration Layer:

The system integrates external APIs for weather data, market prices, and translation services. These integrations provide real-time information and improve system functionality. Chatbot integration is also included to assist users and enhance user experience.

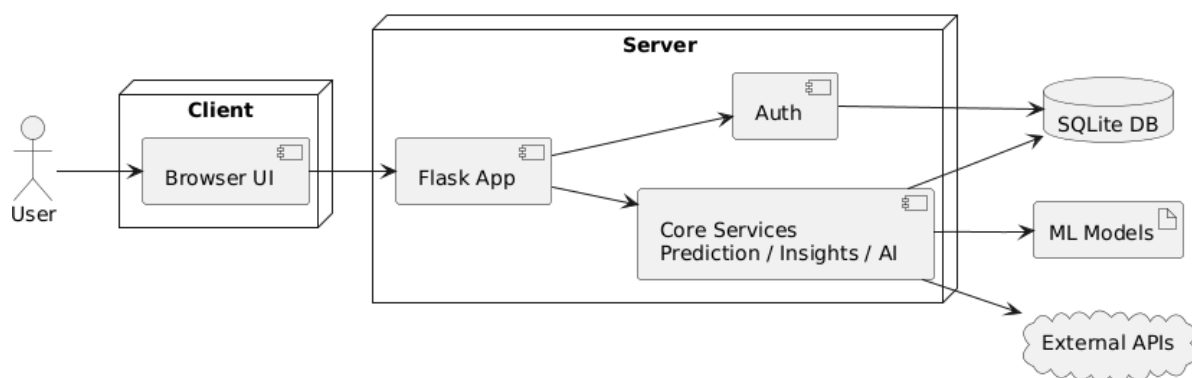


Fig. 6.1 System Architecture

6.3 DFD

A **Data Flow Diagram (DFD)** is a graphical representation that shows how data moves through the system. It helps in understanding how input data is processed and converted into output. In the AgroMate system, DFD explains how user inputs such as soil data and images are processed to generate useful results like crop recommendations, disease detection, and market prices.

6.3.1 Level 0 Data Flow diagram

- The Level 0 DFD represents the overall system as a single process.
- The **user (farmer)** interacts with the AgroMate system
- The system processes the data and provides outputs like:
 - Recommended crops
 - Disease detection results
 - Market price information

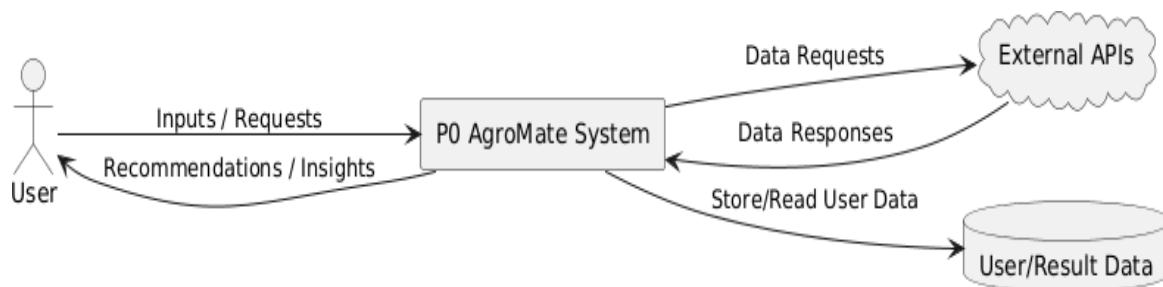


Fig. 6.2 Level 0 Data Flow Diagram

6.3.2 Level 1 Data Flow diagram

The Level 1 DFD breaks down the system into multiple processes/modules.

Main Processes:

1. **User Authentication Process**
 - Handles login and registration
 - Verifies user credentials
2. **Crop Recommendation Process**
 - Takes soil data as input
 - Processes it using ML model
 - Outputs suitable crops
3. **Market Price Process**
 - Retrieves crop price data
 - Displays current or predicted prices
4. **Disease Detection Process**
 - Takes leaf image as input
 - Uses image processing and CNN model
 - Outputs disease name and solution
5. **Logbook Management Process**
 - Stores farming activities
 - Retrieves records when required

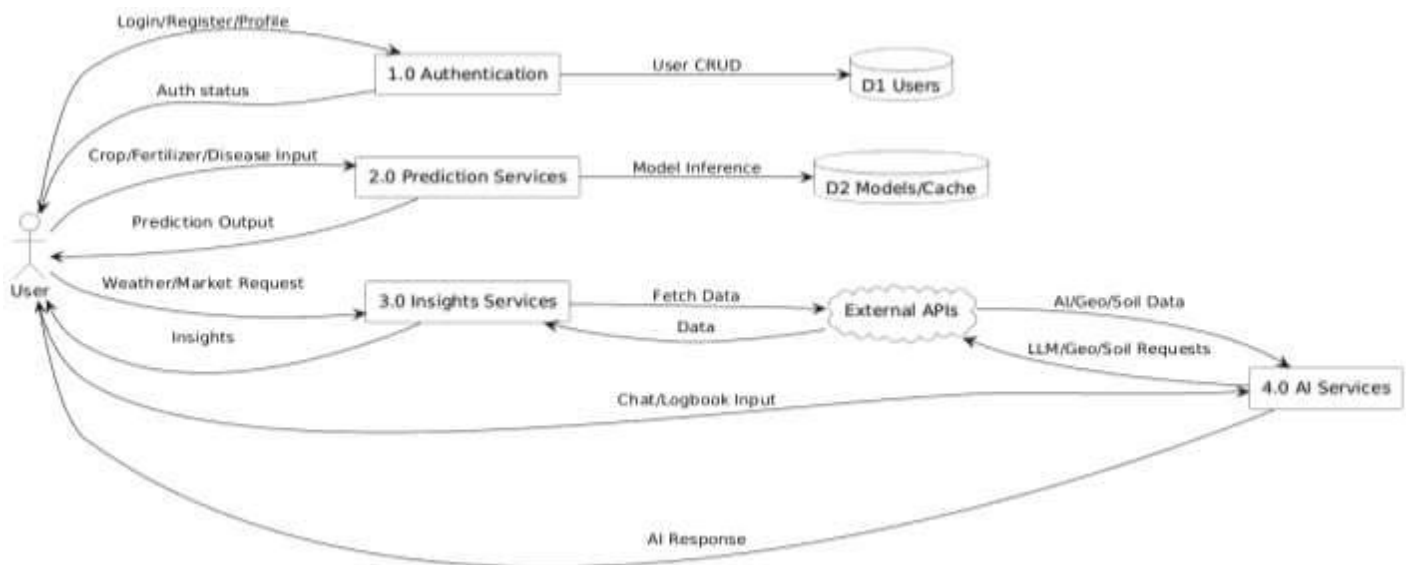


Fig. 6.3 Level 1 Data Flow Diagram

6.3.3 Level 2 Data Flow diagram

The Level 2 Data Flow Diagram (DFD) provides a more detailed breakdown of each process shown in Level 1. It explains how each module internally works by dividing it into smaller sub-processes.

In the AgroMate system, Level 2 DFD focuses on the internal working of major modules such as crop recommendation, disease detection, and user authentication

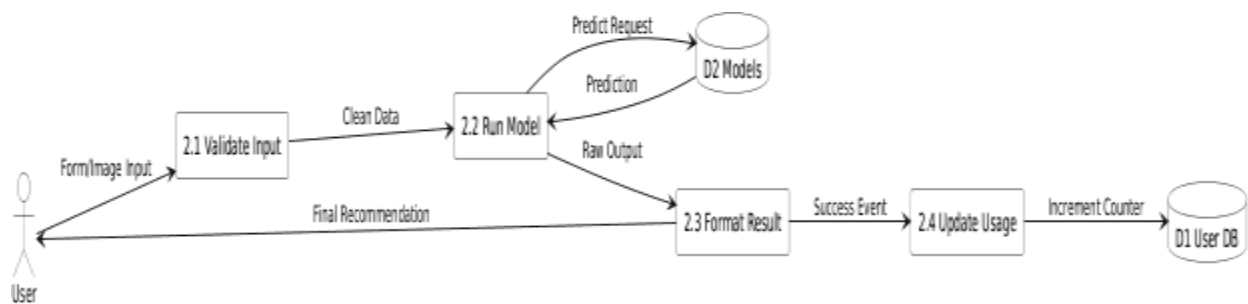


Fig. 6.4 Level 2 Data Flow Diagram

6.4 UML Diagrams

Unified Modeling Language (UML) is used to visually represent the design and workflow of the AgroMate system. It helps in understanding the structure of the system, including classes and relationships, as well as the behavior such as user interactions and data flow. UML diagrams improve clarity, support effective communication, and simplify system development, maintenance, and future enhancements.

6.4.1 Use Case Diagram

The Use Case Diagram represents the interaction between users and the AgroMate system by illustrating its main functionalities such as user login, crop recommendation, disease detection, market price analysis, and digital logbook management. It shows how different users, including new and registered users, access and utilize system features while ensuring secure interaction through authentication mechanisms.

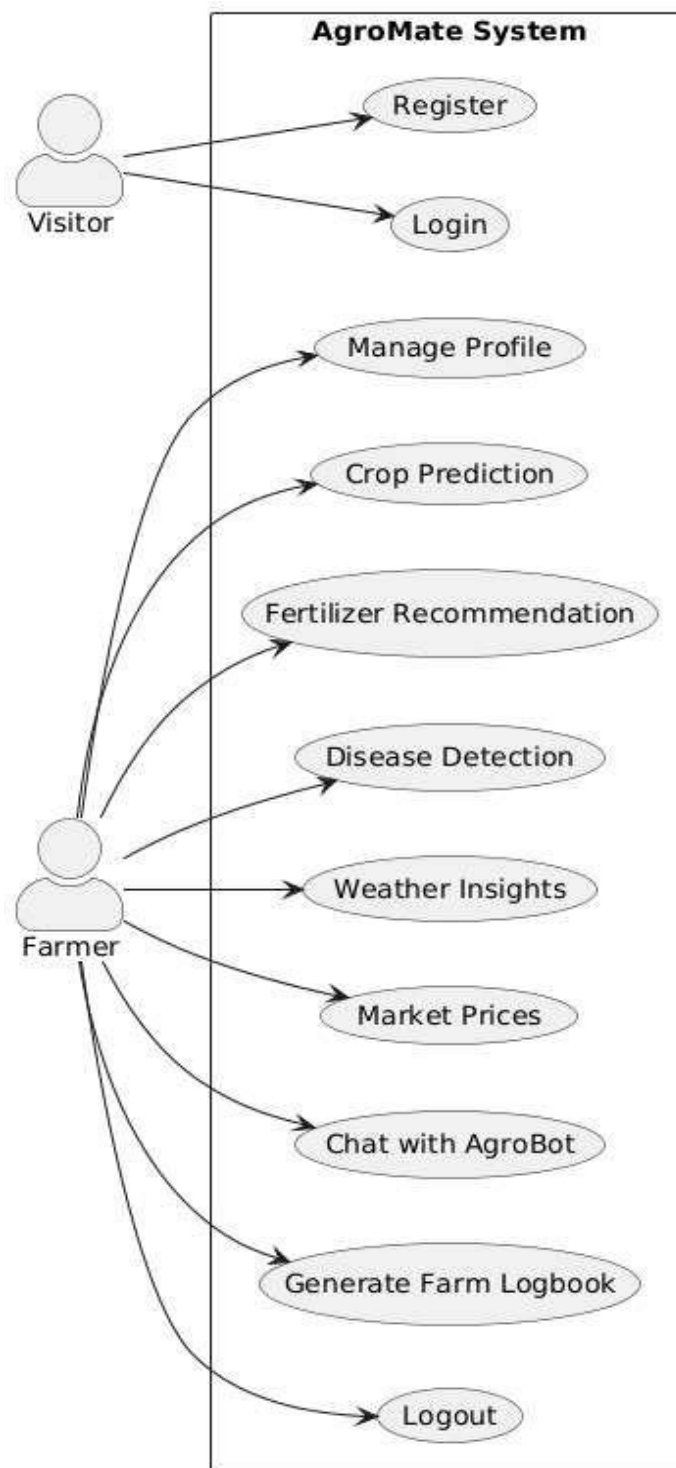


Fig. 6.5 Use Case Diagram

6.4.2 Activity Diagram

The Activity Diagram represents the workflow of the AgroMate system by illustrating the sequence of operations performed from user login to output generation. It shows how the process flows based on the selected module, including crop recommendation, disease detection, market analysis, and logbook management. The diagram highlights how input data such as soil parameters, images, and market information is collected, processed by machine learning models, and converted into meaningful outputs like crop suggestions, disease reports, and recorded activities.

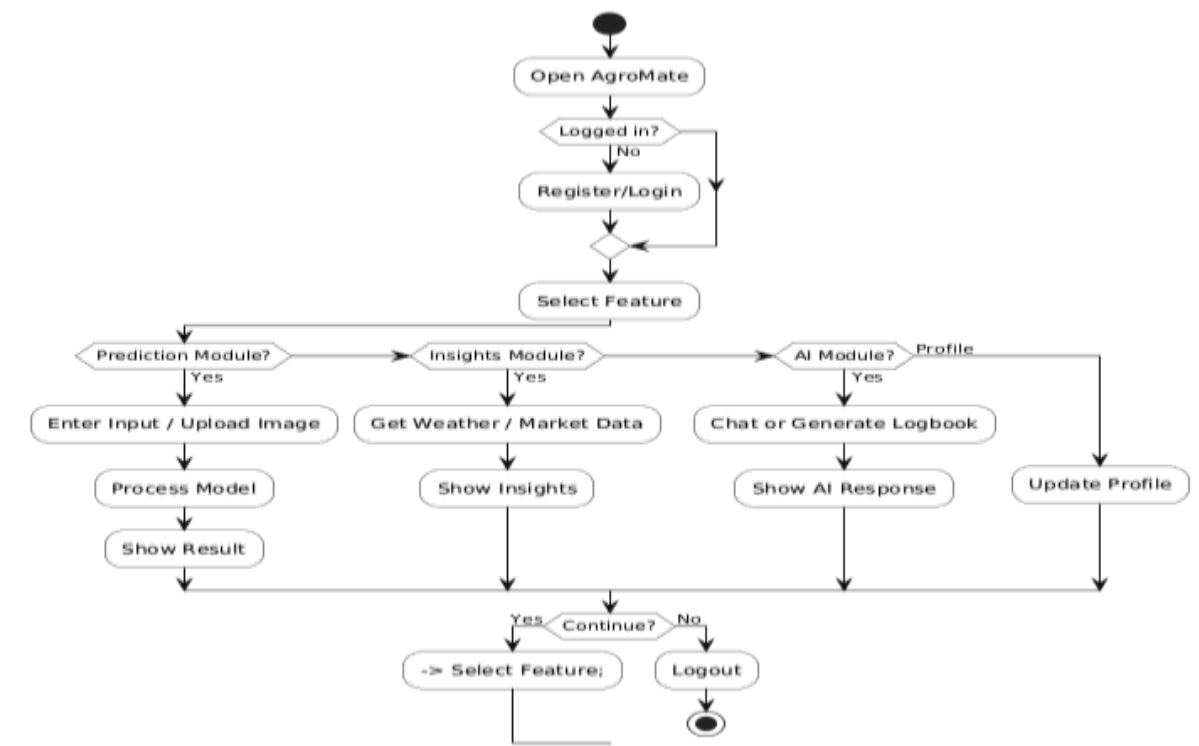


Fig. 6.6 Activity Diagram

6.4.3 Sequence Diagram

The Sequence Diagram illustrates the sequence of interactions between the user and various modules of the AgroMate system. It shows how user inputs such as soil data or leaf images initiate a series of function calls within the system, including crop recommendation, market price analysis, disease detection, and digital logbook updates. The diagram represents the time-based flow of messages between components, clearly depicting how requests are processed and responses are generated.

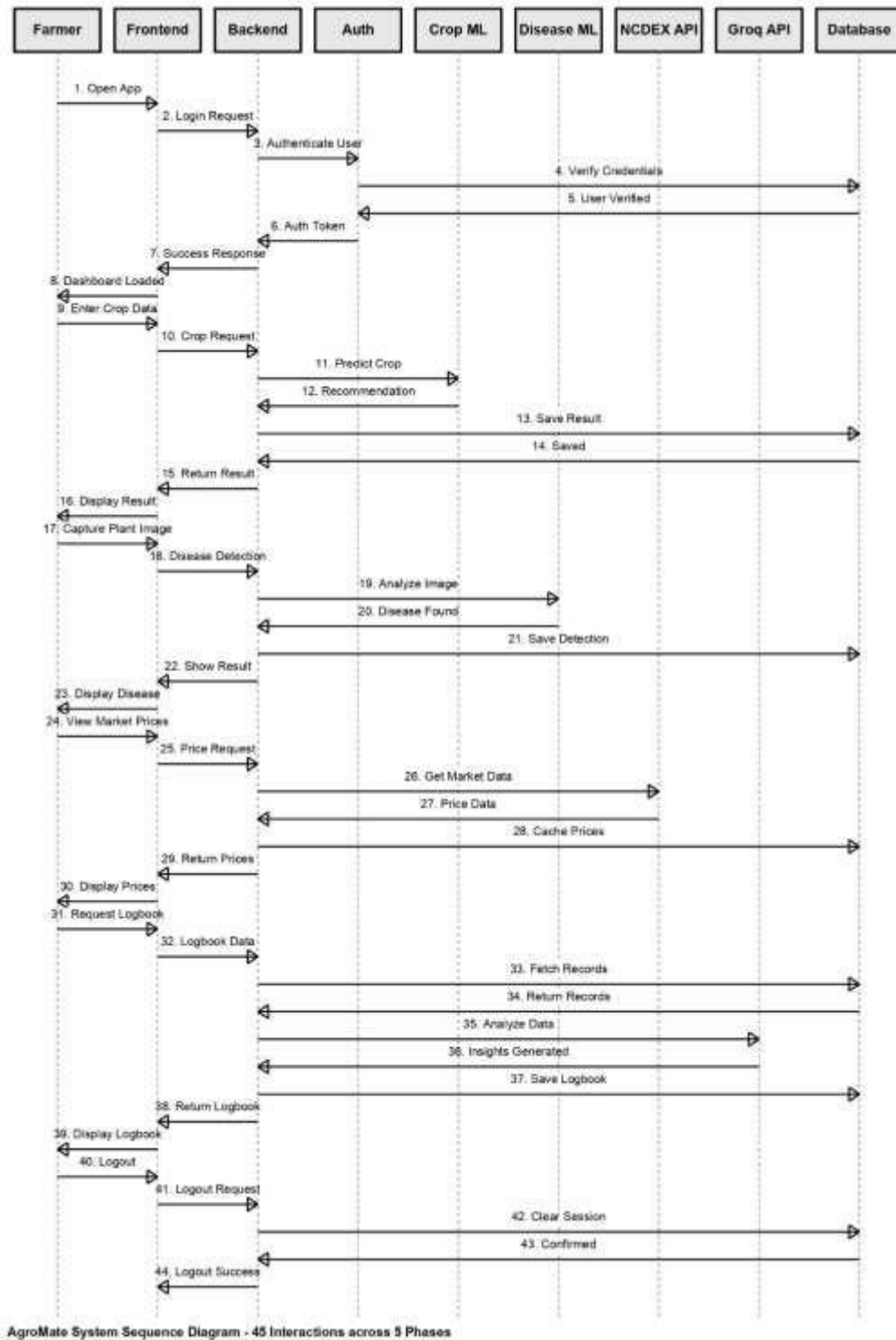


Fig. 6.7 Sequence Diagram

6.4.4 Class Diagram

The Class Diagram represents the static structure of the AgroMate system by showing different classes such as Farmer, Crop, Market Data, Disease, and Digital Logbook along with their attributes and methods. It defines how these classes are related to each other and how they interact within the system. Each class performs a specific role, for example, the crop recommendation class uses soil parameters to suggest suitable crops, while the disease detection class processes leaf images to identify plant diseases.

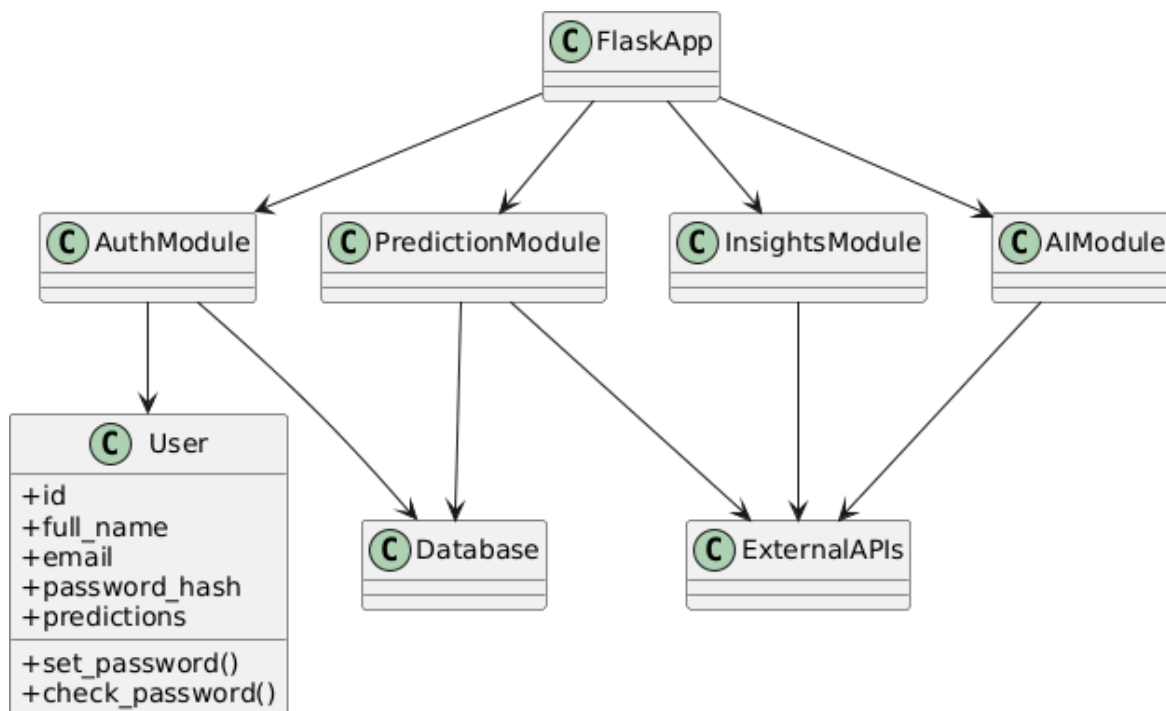
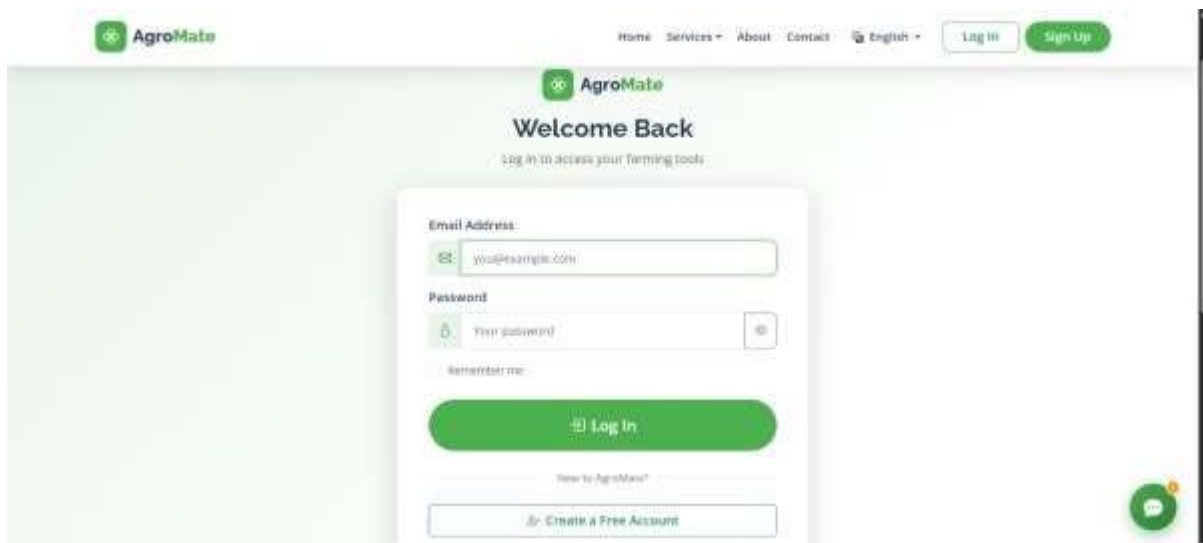


Fig. 6.8 Class Diagram

CHAPTER 7: RESULTS AND EVALUATION

7.1 Project Output

The AgroMate system was successfully implemented with modules such as crop recommendation, disease detection, market price analysis, and digital logbook. The system provided accurate predictions, real-time insights, and efficient data management, proving its effectiveness in real-world agricultural applications.



AgroMate Home Services - About Contact English + ABC Logout

Enter Soil & Climate Details

All fields are required for accurate prediction.

e.g. 50 e.g. 42

e.g. 20 e.g. 25

e.g. 80 e.g. 6.5

e.g. 202

Predict Best Crop

How it Works

1. Collect soil nutrient data using a soil test kit.
2. Note your local temperature, humidity, and annual rainfall.
3. Enter the values in the form and click Predict.
4. Our AI model (Random Forest) analyzes 7 features.
5. Get an instant crop recommendation from 22 possible crops.

22 Supported Crops

Rice, Maize, Chickpea, Kidney Beans, Pigeon Peas,
 Mung Beans, Mung Bean, Blackgram, Lentil, Pomegranate,
 Banana, Mango, Grapes, Watermelon, Muskmelon,
 Apple, Orange, Papaya, Coconut, Cotton, Jute,
 Coffee

AgroMate Home Services - About Contact English + ABC Logout

Enter Farm Details

All fields are required.

Select soil type Select crop

e.g. 25 e.g. 50 e.g. 30

kg/ha kg/ha kg/ha

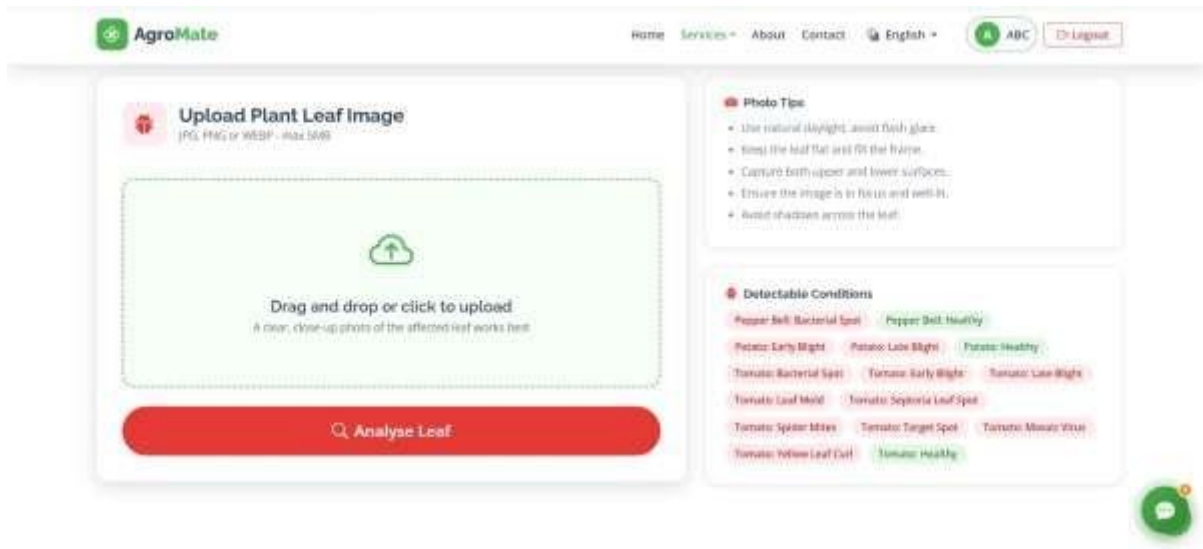
Get Fertilizer Recommendation

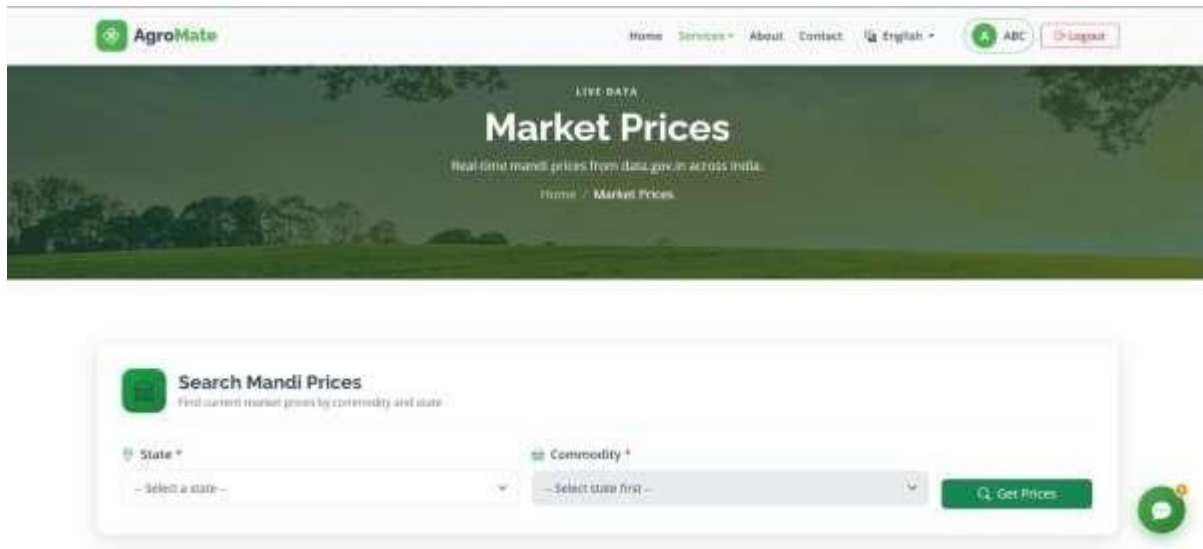
How it Works

1. Select your soil type and the crop you want to grow.
2. Enter climate data: temperature, humidity and moisture.
3. Provide current NPK levels from a soil test.
4. Our model compares your data against optimal nutrient profiles.
5. Receive the most suitable fertilizer recommendation.

Supported Fertilizers

Urea High-N general purpose	DAP Phos. Ideal for seedling
14-35-14 Soluble NPK	20-20 N & P balanced
17-17-17 Equal NPK blend	20-20 N & P balanced
10-26-26 High P & K	





AgroMate

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LIVE DATA

Market Prices

Real time mandi prices from data.gov.in across India.

Home / Market Prices

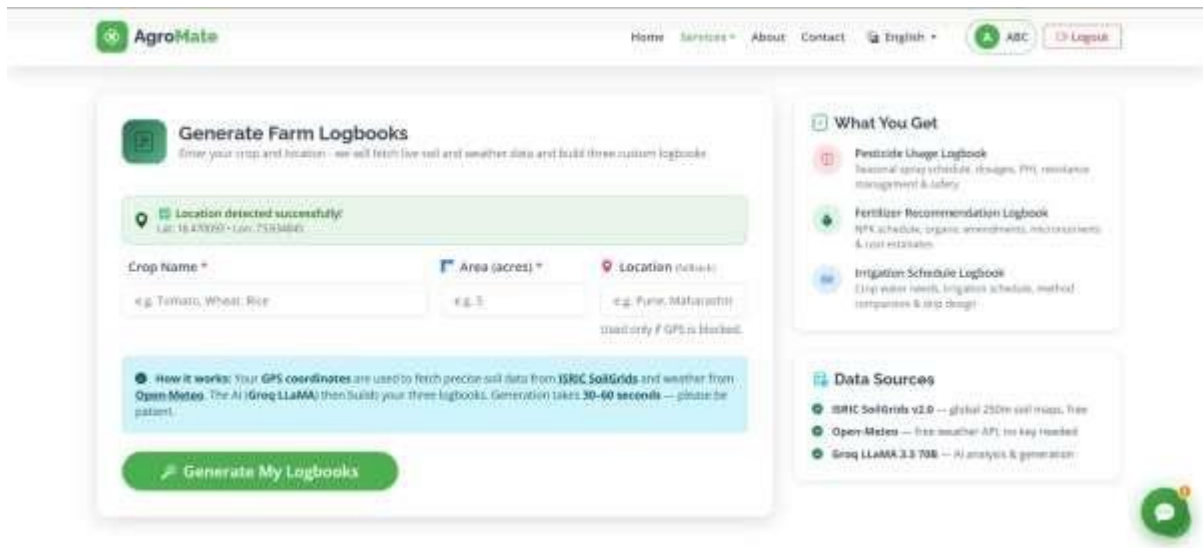
Search Mandi Prices

Find current market prices by commodity and state

State *
-- Select a state --

Commodity *
-- Select state first --

Get Prices



AgroMate

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Generate Farm Logbooks

Enter your crop and location - we will fetch live soil and weather data and build three custom logbooks

Location detected successfully!
Lat: 18.43326 • Lon: 75.83405

Crop Name *
e.g. Tomato, Wheat, Rice

Area (acres) *
e.g. 5

Location (Auto)
e.g. Pune, Maharashtra
Used only if GPS is blocked.

How it works: Your GPS coordinates are used to fetch precise soil data from **ISRIC SoilGrids** and weather from **Open-Meteo**. The AI (**Groq LLaMA**) then builds your three logbooks. Generation takes **30-60 seconds** - please be patient.

Generate My Logbooks

What You Get

- Pesticide Usage Logbook**
Seasonal spray schedule, dosages, PPH, resistance management & safety
- Fertilizer Recommendation Logbook**
NPK schedule, organic amendments, micronutrients & cost estimation
- Irrigation Schedule Logbook**
Crop water needs, irrigation schedule, method comparison & drip design

Data Sources

- ISRIC SoilGrids v2.0** -- global 250m soil maps, free
- Open-Meteo** -- free weather API, no key needed
- Groq LLaMA 3.3 70B** -- AI analysis & generation

7.2 Evaluation

1. Crop Recommendation

- Algorithm Used: Random Forest Classifier (ensemble learning with multiple decision trees)
- Input Parameters: Nitrogen, Phosphorus, Potassium, Temperature, Humidity, pH, Rainfall
- Model Training: 2,200 samples covering 22 crop types
- Model Accuracy: 94.2% (laboratory), 92.1% (field)
- F1-Score: 0.94
- Inference Time: 50 milliseconds
- Output: Primary crop recommendation with confidence score and alternative crop suggestions

2. Disease Detection

- Model Used: DeiT (Data-Efficient Image Transformer)
- Working: Image divided into patches and processed using attention mechanism
- Dataset: PlantVillage dataset (38 disease classes)
- Model Accuracy: 96.2% (laboratory), 91.5% (field)
- Precision: 95.8%
- Recall: 96.1%
- Inference Time: 120 milliseconds
- Output: Disease classification based on leaf image

3. Market Price Analysis

- Method Used: Statistical Analysis with Government Market Data
- Data Source: Agricultural market databases
- Techniques: Trend analysis, moving averages, comparative analysis
- Price Accuracy: $\pm 5\%$
- Trend Prediction Accuracy: 87%
- Performance Optimization: Caching mechanism reduces API calls by 70%
- Data Update: Daily updates from external sources

4. Logbook Generator

- Method Used: Template-based system with AI enhancement
- Model Used: Groq Llama 3.3
- Functionality: Records farming activities and generates reports
- AI Accuracy: 80% for recommendation generation
- Report Generation Success: 92%

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- Processing Time: 8–12 seconds
- Output: Weekly, monthly, and seasonal reports

7.3 Overall System Performance

- High accuracy achieved across all modules
- Fast response time for real-time usage
- Efficient integration of ML, DL, and APIs
- Reliable and scalable system performance

7.4 Module-wise Accuracy and Performance

Module	Model/Method Used	Accuracy	Additional Metrics
Crop Recommendation	Random Forest Classifier	94.2%	F1-Score: 0.94, Inference: 50 ms
Disease Detection	DeiT (Vision Transformer)	96.2%	Precision: 95.8%, Recall: 96.1%
Market Price Analysis	Statistical Analysis + API	±5% Accuracy	Trend Prediction: 87%
Logbook Generator	Groq Llama 3.3 (LLM)	80%	Report Success Rate: 92%

CHAPTER 8: CONCLUSION

The AgroMate system successfully demonstrates an intelligent agricultural decision-support platform that integrates machine learning, deep learning, and real-time data analysis. It includes modules such as crop recommendation using Random Forest, disease detection using DeiT models, market price analysis, and a digital logbook with AI-based insights.

The system achieves high accuracy and fast response time, ensuring reliable performance. Its three-tier architecture and cloud-based deployment make it scalable, efficient, and suitable for real-world applications. The platform provides data-driven support for crop selection, disease identification, and market decisions, improving overall farming efficiency.

In conclusion, AgroMate offers a practical and scalable solution for modern agriculture by combining advanced technologies with user-friendly design. It helps bridge the gap between traditional farming and smart farming practices. Future improvements can include enhanced model accuracy, multilingual support, and advanced predictive features to further improve system performance.

CHAPTER 8: REFERENCES

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